MINI PROJECT REPORT

CampusSpace

Smart Campus Space Finder & Booking System

"Find it. Book it. Use it."

Project Type: B.Tech CSE Mini Project
Development Platform: Base44 AI Tool
Application URL: https://radical-campus-space-flow.base44.app

Scan to open the CampusSpace application

Student Name	_____
Roll Number	_____
Department	Computer Science and Engineering
Institution	_____
Academic Year	2026-2027

Prepared for academic mini-project evaluation, demonstration, and viva.

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1. Executive Summary

CampusSpace is a smart campus space discovery and booking web application designed to help students and faculty quickly identify suitable classrooms, laboratories, library areas, meeting rooms, discussion rooms, seminar halls, and study zones. The solution combines space discovery, reservation management, QR-based check-in, automatic expiry of unused bookings, occupancy reporting, notifications, analytics, and AI-assisted recommendations in a single platform.

The application is developed using the Base44 AI tool, allowing the project to focus on solution design, database structure, business workflows, user experience, and intelligent features rather than building every infrastructure component manually. The system is designed as a practical campus technology solution and as a demonstrable B.Tech CSE mini project.

Core innovation: CampusSpace introduces the SpaceMatch Score, which ranks spaces according to user requirements such as capacity, facilities, current occupancy, time availability, and preferences. This turns a conventional room-booking system into a recommendation-oriented campus platform.

Project URL: https://radical-campus-space-flow.base44.app

2. Abstract

In colleges and universities, students often depend on informal communication channels or manually visit buildings to find suitable places for study, group discussion, project work, or academic activities. This results in wasted time, room conflicts, underutilized spaces, and poor visibility into actual campus resource usage. CampusSpace addresses this problem through a centralized web application that provides space discovery, real-time-style availability views, booking, QR validation, automatic no-show handling, occupancy reporting, notifications, and utilization analytics.

The project demonstrates how AI-assisted application generation can be combined with conventional software engineering practices. Base44 is used as the application development platform, while the design still follows standard principles such as role-based access control, relational data modeling, input validation, deterministic booking rules, workflow automation, and responsive UI design. The result is intended to provide a practical, scalable prototype that could be adapted to a real institutional environment.

3. Introduction

3.1 Background

Campus facilities are shared resources. During peak academic hours, spaces may be difficult to locate, while other rooms may remain unused because students do not know they are available. Traditional room booking processes often lack a single view of availability, do not handle no-shows efficiently, and provide little data for administrators to optimize utilization.

CampusSpace proposes a digital layer over campus spaces. Students can search for a suitable space before walking to it, faculty can reserve rooms for academic use, and administrators can monitor patterns of demand and utilization. The system is designed to reduce friction in everyday campus activities while creating useful operational data.

3.2 Need for the Project

- Reduce the time spent searching for suitable campus spaces.
- Prevent overlapping reservations and improve booking transparency.
- Release unused reservations through no-show handling and automatic expiry.

- Provide a consistent way to report occupancy information.
- Help administrators understand which spaces are popular or underused.
- Provide intelligent recommendations instead of forcing users to browse every room.

4. Problem Statement

Students waste time searching for available classrooms, labs, library seats, seminar halls, and discussion spaces. Information may be scattered across notices, messaging groups, manual registers, or verbal communication. Even when a room is technically available, users may not know whether it is appropriate for their group size, preferred facilities, or study purpose. At the administrative level, the absence of a unified data source makes utilization analysis difficult.

Proposed statement: Design and implement a web-based smart campus space platform that allows authorized users to discover suitable spaces, check availability, reserve them without conflicts, validate occupancy through QR-based check-in, automatically release unused bookings, and provide administrators with analytics for campus resource utilization.

5. Objectives

1. Develop a centralized campus space discovery and booking application.
2. Provide secure login and role-based access for students, faculty/staff, and administrators.
3. Enable advanced search using time, capacity, building, facilities, and space type.
4. Provide AI-assisted recommendations through the SpaceMatch Score.
5. Implement booking validation that prevents overlapping reservations.
6. Introduce QR-based check-in for booking verification.
7. Automatically expire unused bookings after a configurable grace period.
8. Collect occupancy reports and present utilization analytics.
9. Send useful notifications for booking-related events.
10. Build a responsive and professional user experience suitable for a real campus environment.

6. Scope of the Project

6.1 In Scope

- Campus, building, floor, and space management.
- Student and faculty booking workflows.
- Admin management and analytics dashboard.
- Search and filtering of available spaces.
- AI-assisted natural-language space recommendation.
- QR-based booking check-in.
- Automatic no-show release and booking completion.
- Occupancy reporting and notifications.
- Responsive web application interface.

6.2 Out of Scope

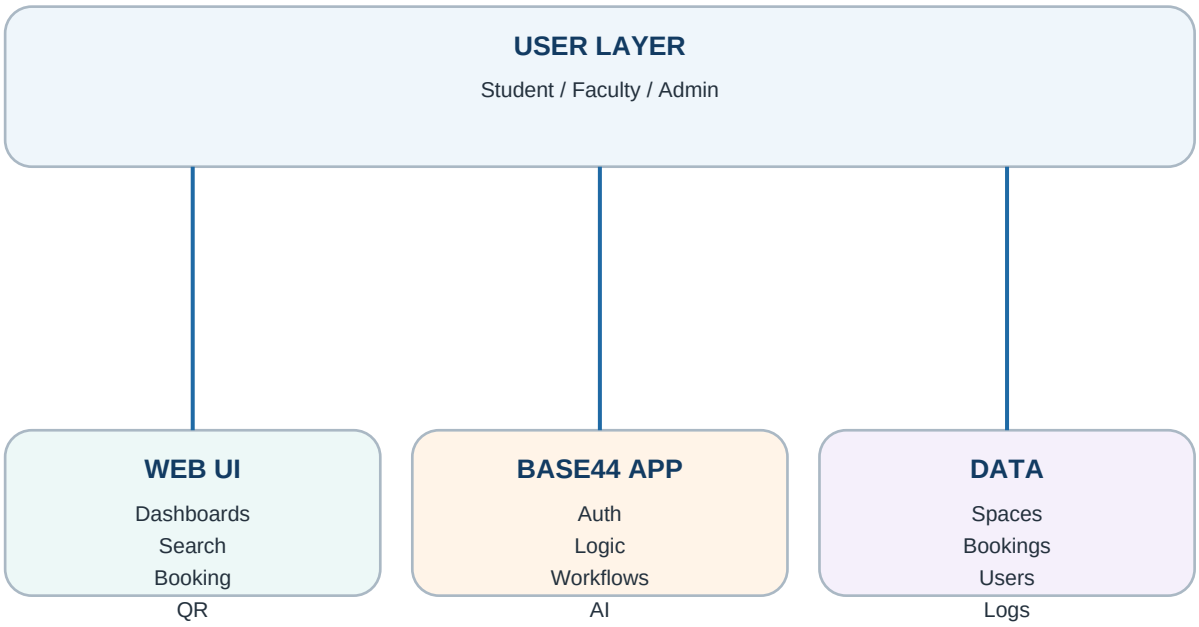
- Physical IoT occupancy sensors and hardware installation.
- Official university identity integration unless an institution provides the required service.
- Native Android/iOS applications; this project focuses on the responsive web application.
- Financial payment processing for facilities.
- Guaranteed real-world occupancy measurement without trusted sensor data.

7. Existing System vs Proposed System

Parameter	Existing Approach	CampusSpace
Space discovery	Manual search, notices, or messages	Centralized search with filters and recommendations
Availability	May be outdated or scattered	Space-level availability with booking state
Reservation	Manual register or informal requests	Structured booking workflow
Conflict control	Human verification	System validation against overlapping bookings
No-show handling	Usually manual	Grace period and automatic release
Occupancy insight	Limited	Reports, trends, and analytics
Intelligence	Little or none	AI-assisted SpaceMatch recommendations
Check-in	Manual	QR-based validation

8. System Architecture

CampusSpace follows a layered web-application architecture. The user interface is responsible for navigation, data presentation, forms, dashboards, and interaction. Base44 provides the application layer for authentication, data access, business logic, workflows, and AI-assisted functions. The data layer stores users, campuses, buildings, spaces, bookings, occupancy reports, notifications, QR codes, and logs.



Architecture flow: User -> Responsive Web UI -> Base44 application logic and workflows -> Persistent application data. AI-assisted recommendations operate on top of structured space and booking data and do not replace deterministic reservation validation.

9. Technology Stack

Layer	Technology / Approach	Purpose
Application platform	Base44 AI Tool	Rapid full-stack application generation and hosting
Frontend	Responsive web UI generated within the platform	Dashboards, forms, search, booking, notifications
Backend logic	Base44 application logic / workflows	Validation, booking rules, scheduled tasks
Database	Base44-managed application data	Users, spaces, bookings, reports, logs
AI	Base44 AI capabilities	Natural-language recommendation and explanation
QR	QR code generation/validation flow	Space-level check-in
Visualization	Dashboard cards and charts	Utilization analytics

10. Requirements Analysis

10.1 Functional Requirements

1. The system shall support user registration and login.
2. The system shall assign access according to user role.
3. Users shall be able to search and filter campus spaces.
4. Users shall be able to view space details and availability.
5. Users shall be able to create and cancel bookings according to policy.

6 The system shall reject overlapping or invalid bookings.

7 The system shall generate a unique booking identifier.

8 The system shall validate QR-based check-in against booking and time conditions.

9 The system shall mark no-shows and release unused reservations automatically.

10 The system shall store and display occupancy reports.

11 The system shall send or display booking-related notifications.

12 Administrators shall be able to manage spaces and view utilization analytics.

10.2 Non-Functional Requirements

- **Usability:** Interfaces should be easy to understand for students and staff.
- **Security:** Access control must protect admin functions and user-owned booking data.
- **Reliability:** Booking logic must consistently prevent conflicts.
- **Performance:** Common tasks such as search and viewing bookings should feel responsive.
- **Scalability:** The design should support additional buildings and spaces.
- **Maintainability:** Business rules and entities should remain logically separated.
- **Responsiveness:** The application should work on desktop, tablet, and mobile screens.

11. Major Modules

11.1 Authentication and User Management

Handles signup, login, sessions, profile information, and role-based authorization for Student, Faculty/Staff, and Admin users.

11.2 Space Discovery

Provides search, filtering, sorting, detailed room information, availability status, and facilities.

11.3 Smart Recommendation

Interprets user requirements and produces ranked recommendations using SpaceMatch Score and supporting explanations.

11.4 Booking Management

Creates, validates, confirms, cancels, completes, and tracks reservations without allowing overlapping bookings.

11.5 QR Check-in

Validates a user's active reservation against the space QR code and records check-in time.

11.6 Occupancy Reporting

Collects user-reported occupancy information and provides a moderation workflow for administrators.

11.7 Notification Center

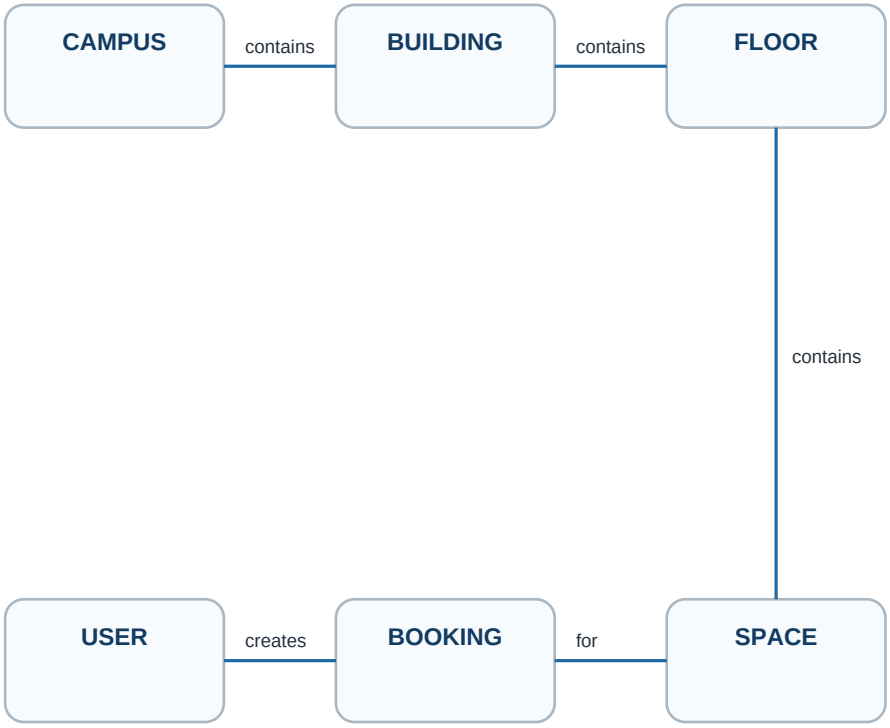
Provides booking confirmations, reminders, cancellations, expiry alerts, room notices, and admin announcements.

11.8 Admin and Analytics

Provides management functions for spaces, users, bookings, policies, dashboards, charts, and campus utilization insights.

12. Database Design

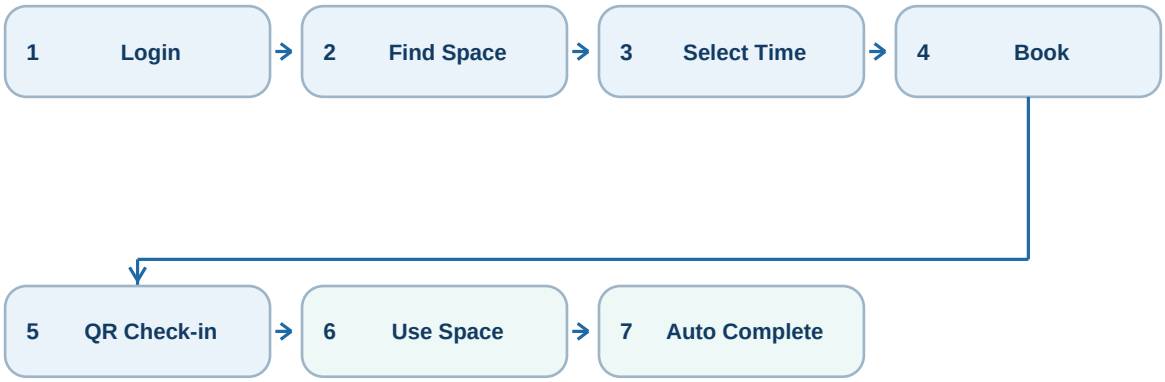
The logical data model is organized around a campus hierarchy and reservation lifecycle. The core hierarchy is Campus -> Building -> Floor -> Space. User and Booking records connect people to space reservations, while OccupancyReport, Notification, QRCode, and AdminActionLog support operational features.



Entity	Important Attributes	Purpose
User	id, name, email, role, department, profileImage	Stores account and role information
Campus	id, name, location, description	Represents a campus location
Building	id, campusId, name, code	Groups floors and spaces
Floor	id, buildingId, floorNumber, name	Organizes spaces by level
Space	id, floorId, name, code, type, capacity, facilities, status	Represents a bookable or visible campus space
Booking	id, userId, spaceId, date, startTime, endTime, purpose, status, checkInStatus	Stores reservation lifecycle
OccupancyReport	id, spaceId, userId, reportedOccupancy, status, createdAt	Stores occupancy observations
Notification	id, userId, title, message, type, readStatus	Stores user notifications
QRCode	id, spaceId, uniqueCode, active	Maps QR identifiers to spaces
AdminActionLog	id, adminId, action, targetType, targetId, timestamp	Audits important admin activity

13. System Workflow

The complete student workflow is designed to minimize friction. The user first logs in, identifies the needed space, checks availability, completes a validated reservation, and later confirms actual use by scanning the space QR code. The system then closes the reservation automatically when the time window ends.



13.1 Booking Workflow

1. User selects a space and enters date, start time, end time, purpose, and group size.
2. The system checks operating hours, capacity, duration limits, and existing overlapping bookings.
3. A booking summary is displayed for confirmation.
4. On confirmation, a unique booking record is created.
5. A notification is generated and the booking appears in My Bookings.
6. At the start time, the user may check in using the space QR code.
7. If the user does not check in within the configured grace period, the system can mark the reservation as No Show and release it.
8. At the end time, an active booking is completed automatically.

13.2 Smart Recommendation Workflow

1. User describes the need in natural language or selects structured filters.
2. The system extracts requirements such as group size, duration, facilities, and purpose.
3. Candidate spaces are checked for availability and policy constraints.
4. Eligible spaces are ranked using SpaceMatch Score.
5. The user sees recommendations together with concise reasons for each match.
6. The user selects a recommended space and continues through the normal deterministic booking flow.

14. User Interface and Navigation

14.1 Student Navigation

Dashboard | *Find Space* | *My Bookings* | *Notifications* | *Profile*

14.2 Admin Navigation

Dashboard | *Spaces* | *Buildings* | *Bookings* | *Users* | *Occupancy* | *Analytics* | *Notifications* | *Settings*

14.3 Key Screens

- **Landing Page:** Product introduction, benefits, CTA buttons, and workflow overview.
- **Student Dashboard:** Availability, upcoming bookings, quick actions, and summary statistics.
- **Find Space:** Search bar, filters, sorting, and smart recommendations.
- **Space Details:** Capacity, occupancy, facilities, rules, calendar, and booking action.
- **Booking Confirmation:** Summary, booking ID, and post-booking status.
- **My Bookings:** Upcoming and history views with status and actions.
- **Admin Dashboard:** KPIs, charts, utilization trends, and operational insights.

15. Implementation Using Base44 AI Tool

Base44 is used as the primary application development environment. The project prompt specifies the user roles, data entities, booking logic, workflows, AI recommendation behavior, and page structure so that the generated application is aligned with the required system design. The project is then refined through iterative instructions that improve visual consistency, error handling, responsiveness, and functional behavior.

15.1 Why Base44 Was Chosen

- Allows rapid prototyping of a full-stack web application from natural-language requirements.
- Supports iterative development without requiring every UI and backend component to be manually assembled.
- Provides a suitable environment for authentication, data-driven pages, application workflows, and AI-assisted features.
- Makes it practical to demonstrate a working product within mini-project time constraints.

15.2 Development Method

1. Define the problem statement and core user journeys.
2. Write a detailed single master prompt covering functionality, data, security, and UI.
3. Generate the initial application in Base44.
4. Review pages, workflows, database behavior, and role permissions.
5. Refine the application using targeted prompts for design and reliability improvements.
6. Load realistic demo data for project presentation.
7. Test the major user and admin flows and fix issues discovered during review.

16. Innovative Features

Feature	Innovation / Benefit
SpaceMatch Score	Ranks suitable spaces according to capacity, facilities, occupancy, availability, and user preferences.
Natural-language search	Lets users express needs in ordinary language such as "quiet room for four with projector".
QR check-in	Links physical space use to a valid digital reservation.
Automatic no-show release	Returns unused capacity to the pool instead of leaving rooms unnecessarily locked.
Occupancy reporting	Allows the community to contribute observations while preserving distinction between reported and sensor-verified data.
Utilization analytics	Helps administrators identify demand patterns, peak times, and underused resources.

17. Testing Strategy

Testing should combine functional checks, workflow checks, authorization checks, and responsive UI checks. Because booking is the critical business rule, conflict validation and edge cases deserve special attention.

Test Case	Expected Result	Status
Valid login	User enters the appropriate dashboard	Pass / Verify
Invalid login	Friendly authentication error is shown	Pass / Verify
Valid booking	Booking created with unique ID	Pass / Verify
Overlapping booking	Second booking is rejected	Pass / Verify
Outside operating hours	Booking is rejected	Pass / Verify
Over-capacity booking	Booking is rejected	Pass / Verify
Valid QR check-in	Check-in recorded for the correct booking and space	Pass / Verify
Invalid QR check-in	Friendly rejection shown	Pass / Verify
No-show booking	Booking becomes No Show and slot is released after grace period	Pass / Verify
Unauthorized admin route	Access is blocked	Pass / Verify
Mobile viewport	Navigation and booking flow remain usable	Pass / Verify

18. Expected Benefits

- Students spend less time searching for study and project spaces.
- Faculty can reserve suitable rooms more systematically.
- The campus can reduce space wastage caused by no-shows.
- Administrators receive better visibility into demand and utilization.
- The application creates a foundation for future smart-campus extensions.
- The project demonstrates practical application of AI-assisted software development and software engineering principles.

19. Limitations

- Actual occupancy accuracy depends on the availability and reliability of campus data; user reports are not equivalent to physical sensors.
- Production deployment may require institutional identity integration, data governance, and security review.
- Real-time behavior depends on the capabilities and configuration available in the deployed application environment.
- Email or external integrations may require additional configuration and permissions.
- The project prototype should be validated against the institution's actual booking policies before operational use.

20. Future Enhancements

1. Integrate IoT sensors for trusted occupancy measurement.
2. Add indoor campus maps and floor-level navigation.
3. Introduce QR-based room entry with institutional identity integration.
4. Add calendar integration for Google Calendar or Microsoft 365.
5. Build native mobile applications for Android and iOS.
6. Use historical demand to forecast peak periods and suggest alternate spaces.
7. Add accessibility-aware recommendations and route planning.
8. Introduce institution-wide reporting and export capabilities.
9. Add multilingual support for campus users.

21. Conclusion

CampusSpace presents a practical solution to a common campus problem: finding and using shared spaces efficiently. By combining space discovery, booking, QR check-in, automated no-show release, occupancy reporting, notifications, and analytics, the application provides value to both end users and administrators. The SpaceMatch Score adds an intelligent recommendation layer that makes the platform more than a simple reservation system.

Using Base44 AI as the development platform enables rapid creation and refinement of the application while still allowing the project to demonstrate important CSE concepts such as authentication, authorization, data modeling, business rules, workflow automation, user-centered design, testing, and AI-assisted functionality. The project can serve as a strong mini-project prototype and a foundation for future smart-campus capabilities.

22. Project Reference and Access

Application / Demo URL: https://radical-campus-space-flow.base44.app

The above link points to the CampusSpace web application generated and deployed through Base44. For a final institutional submission, replace the placeholder student/institution details on the cover page with the actual project information.

22.1 Recommended GitHub Post Content

CampusSpace - Smart Campus Space Finder & Booking System is a B.Tech CSE mini project built using the Base44 AI tool. It helps students and faculty discover suitable campus spaces, check availability, book rooms, verify usage through QR check-in, and benefit from smart recommendations using SpaceMatch Score. The project also includes automatic no-show release, occupancy reporting, notifications, and administrative analytics. Live application: https://radical-campus-space-flow.base44.app

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End of Mini Project Documentation